

PART A

Explain the terms emitter injection efficiency and base transport factor of a BJT. (3)

6 Differentiate Schottky and rectifying contacts metal-Semiconductor contacts. (3)

5)

The total emitter current i_E is made up of the hole component i_{Ep} and the electron component i_{En} , due to electrons injected from the base to emitter

The emitter injection efficiency γ is

$$\gamma = \frac{i_{Ep}}{i_{En} + i_{Ep}}$$

base. Thus i_C is proportional to the hole component of the emitter current i_{Ep} :

$$i_C = Bi_{Ep}$$

The proportionality factor B is simply the fraction of injected holes which make it across the base to the collector; B is called the *base transport factor*.

PART A

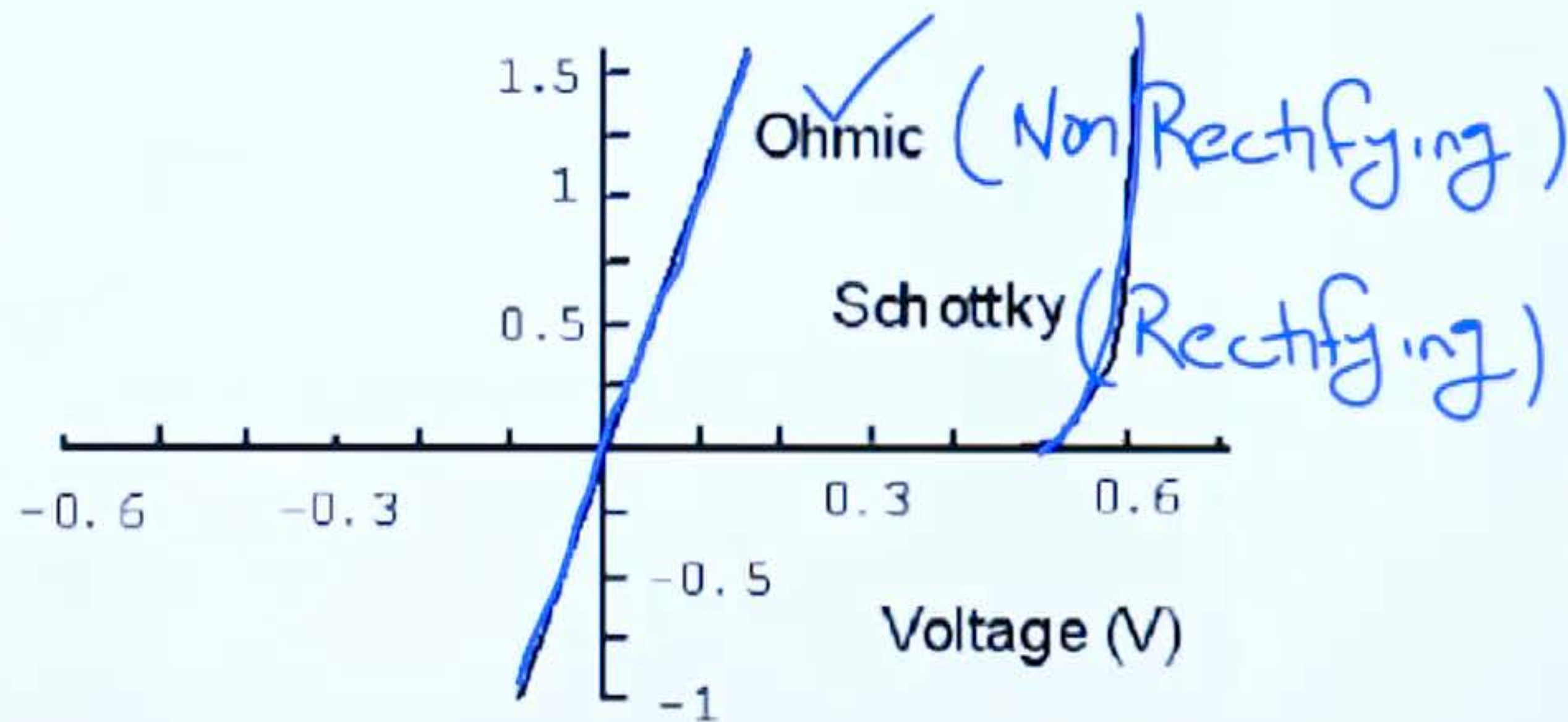
- 5 Explain the terms emitter injection efficiency and base transport factor of a BJT. (3)
- 6 Differentiate Ohmic and rectifying contacts metal-Semiconductor contacts. (3)

6)



When you create a junction between a metal and a semiconductor, some interesting things can happen. An ohmic contact is a metal-semiconductor junction that allows current to flow both ways roughly equally within normal device operation range, with a voltage-current relationship (I-V curve) that comes close to that of a resistor (hence the name "ohmic"). A rectifying contact only allows current to flow one way, with an I-V curve that looks a lot like a diode -- in fact, it's usually referred to as a Schottky diode. (Hence the name "rectifying," as in a rectifier circuit to convert AC to DC).

Current density (kA/cm^2)



Type of contact	Relative work function	
	$\phi_m > \phi_{sc}$	$\phi_m < \phi_{sc}$
<u>Metal N-type S.C</u>	<u>Rectifying</u>	<u>Ohmic</u>
<u>Metal P-type S.C</u>	<u>Ohmic</u>	<u>Rectifying</u>



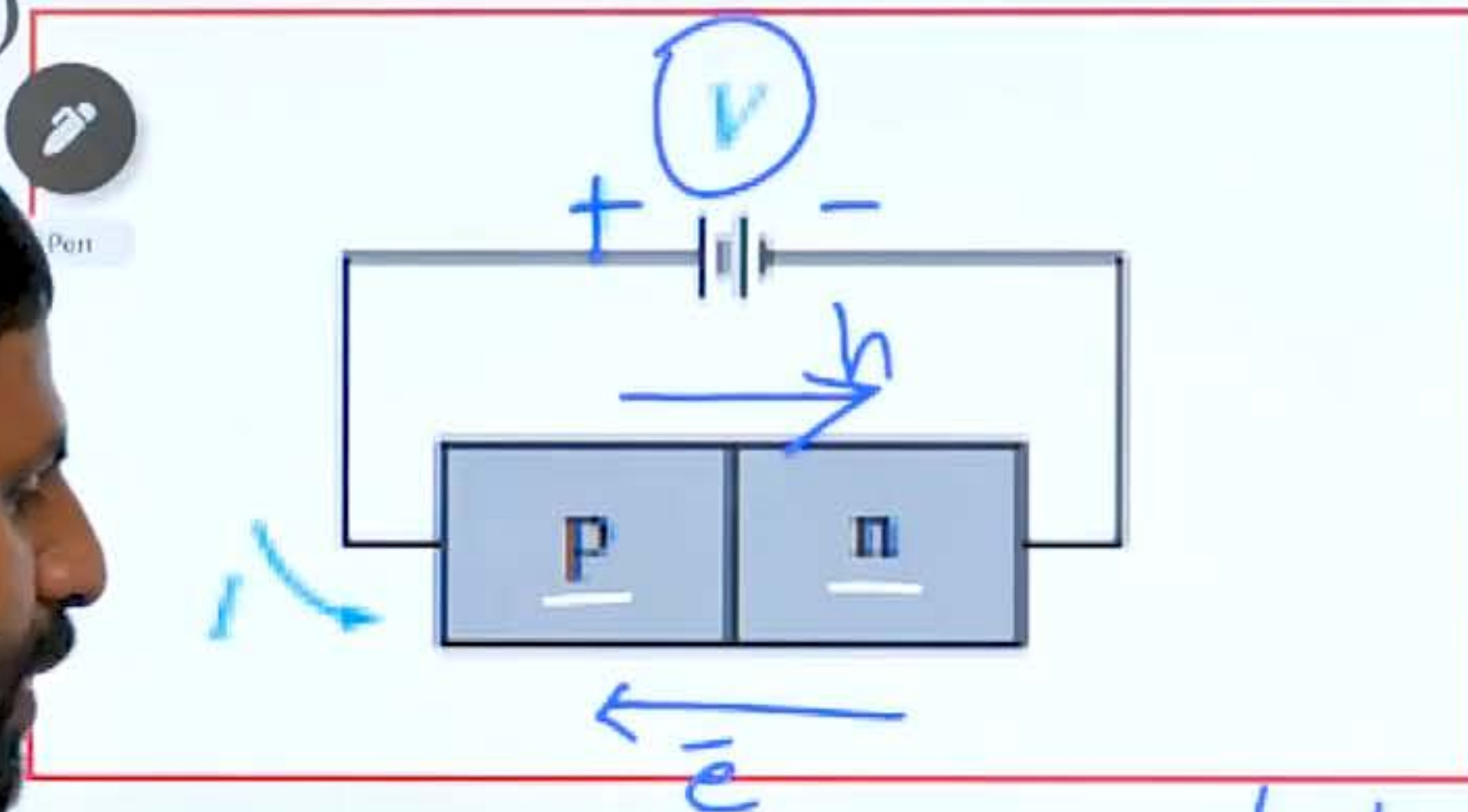
Pen

15 (a) Derive ideal diode equation. State any two assumptions used. (8)

(b) Draw the energy band diagram of a metal-n type semiconductor with $\phi_m > \phi_s$ when it is i) under equilibrium and ii) when it is biased. Is the contact rectifying or ohmic? (6)



15 a)



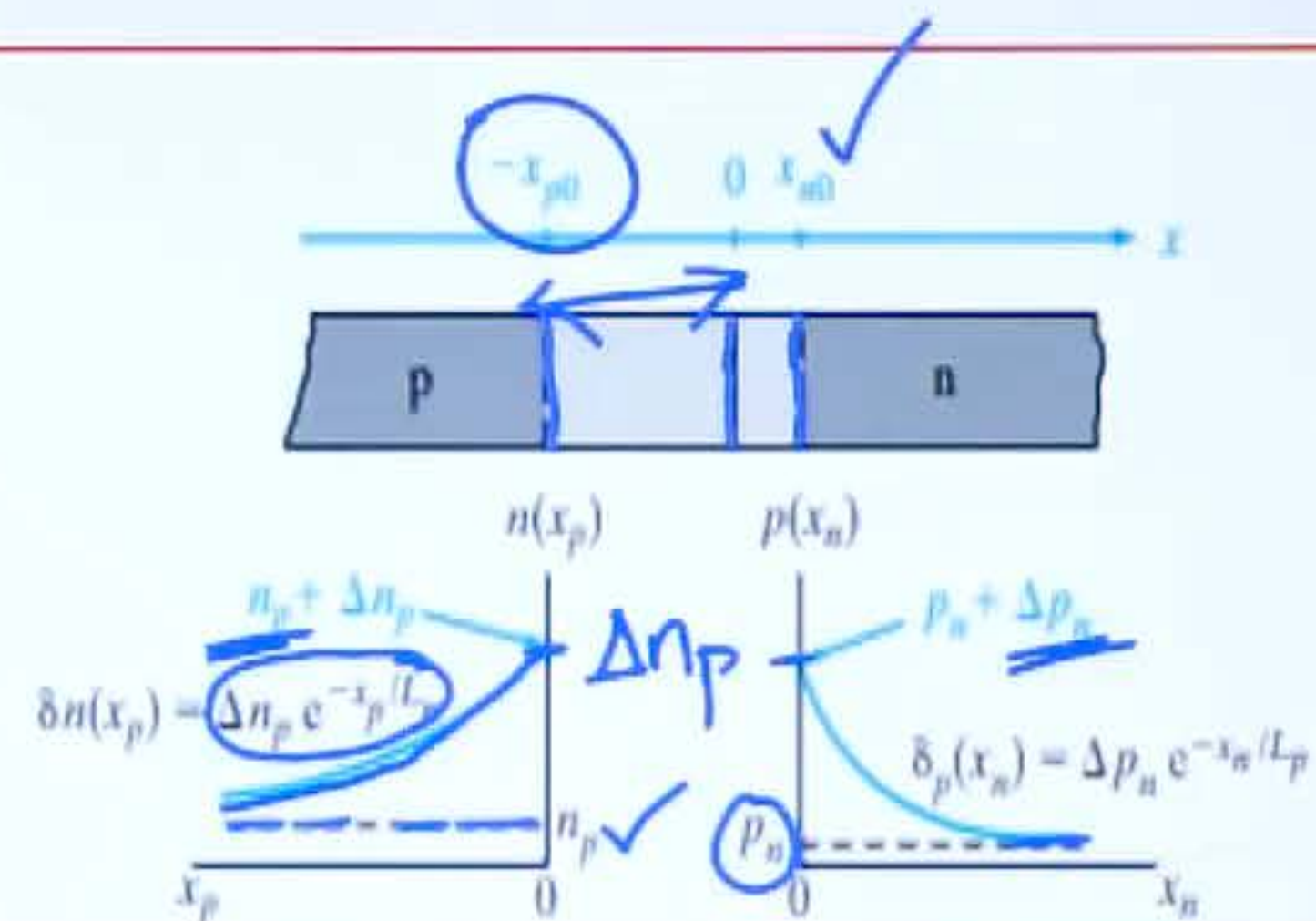
$(V_0 - V)$

At equilibrium

$$\frac{p_p}{p_n} = e^{qV_0/kT}$$

With bias

$$\frac{p(-x_{p0})}{p(x_{n0})} = e^{q(V_0 - V)/kT}$$



$$\Delta n_p e^{-x_p/L_n}$$

$$\Delta p_n e^{-x_n/L_p}$$

excess concentration is

$$\checkmark \Delta p_n = p(x_{n0}) - p_n = p_n(e^{qV/kT} - 1)$$

$$\checkmark \Delta n_p = n(-x_{p0}) - n_p = n_p(e^{qV/kT} - 1)$$

excess carrier distribution is

$$\rightarrow \delta n(x_p) = \Delta n_p e^{-x_p/L_n} = n_p(e^{qV/kT} - 1)e^{-x_p/L_n}$$

$$\rightarrow \delta p(x_n) = \Delta p_n e^{-x_n/L_p} = p_n(e^{qV/kT} - 1)e^{-x_n/L_p}$$

The hole diffusion current at any point x_n in the n material can be calculated from Eq. (4-40):

$$\underline{I_p(x_n)} = -qAD_p \frac{d\delta p(x_n)}{dx_n} = qA \frac{D_p}{L_p} \Delta p_n e^{-x_n/L_p} = qA \frac{D_p}{L_p} \delta p(x_n) \quad (5-32)$$

$$\underline{I_p(x_n = 0)} = \frac{qAD_p}{L_p} \underline{\Delta p_n} = \frac{qAD_p}{L_p} p_n (e^{qV/kT} - 1) \quad \checkmark$$

Similarly

$$\underline{I_n(x_p = 0)} = -\frac{qAD_n}{L_n} \Delta n_p = -\frac{qAD_n}{L_n} n_p (e^{qV/kT} - 1)$$

$$\underline{I = I_p(x_n = 0) - I_n(x_p = 0) = \frac{qAD_p}{L_p}\Delta p_n + \frac{qAD_n}{L_n}\Delta n_p}$$

$$I = qA\left(\frac{D_p}{L_p}p_n + \frac{D_n}{L_n}n_p\right)(e^{qV/kT} - 1) = I_0(e^{qV/kT} - 1)$$

↓
 I_0/I_s

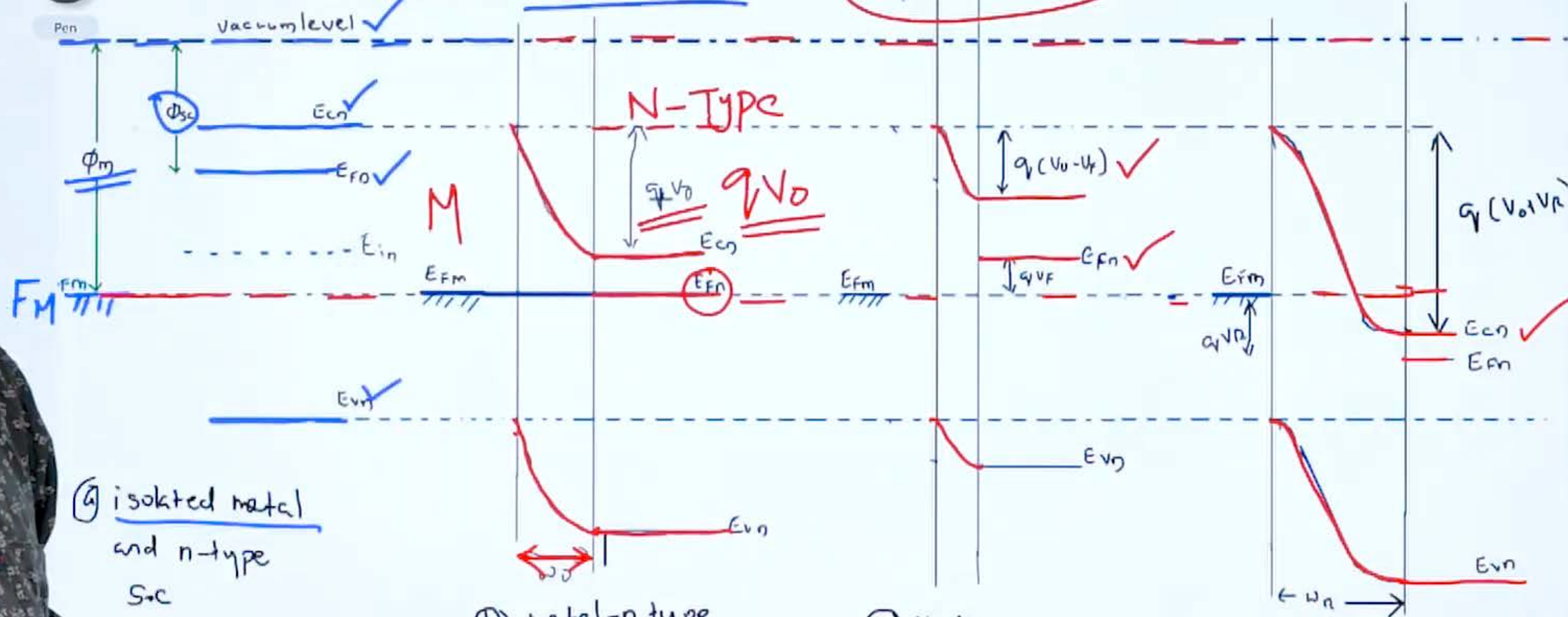
- * length $> L_n, L_p$
- * low level injection

15 b)



MÉTAL - N TYPE SEMICONDUCTOR SCHOTTKY CONTACT

$\phi_m > \phi_s$



(a) isolated metal and n-type semiconductor

(b) metal-n type at equilibrium

(c) Under forward bias

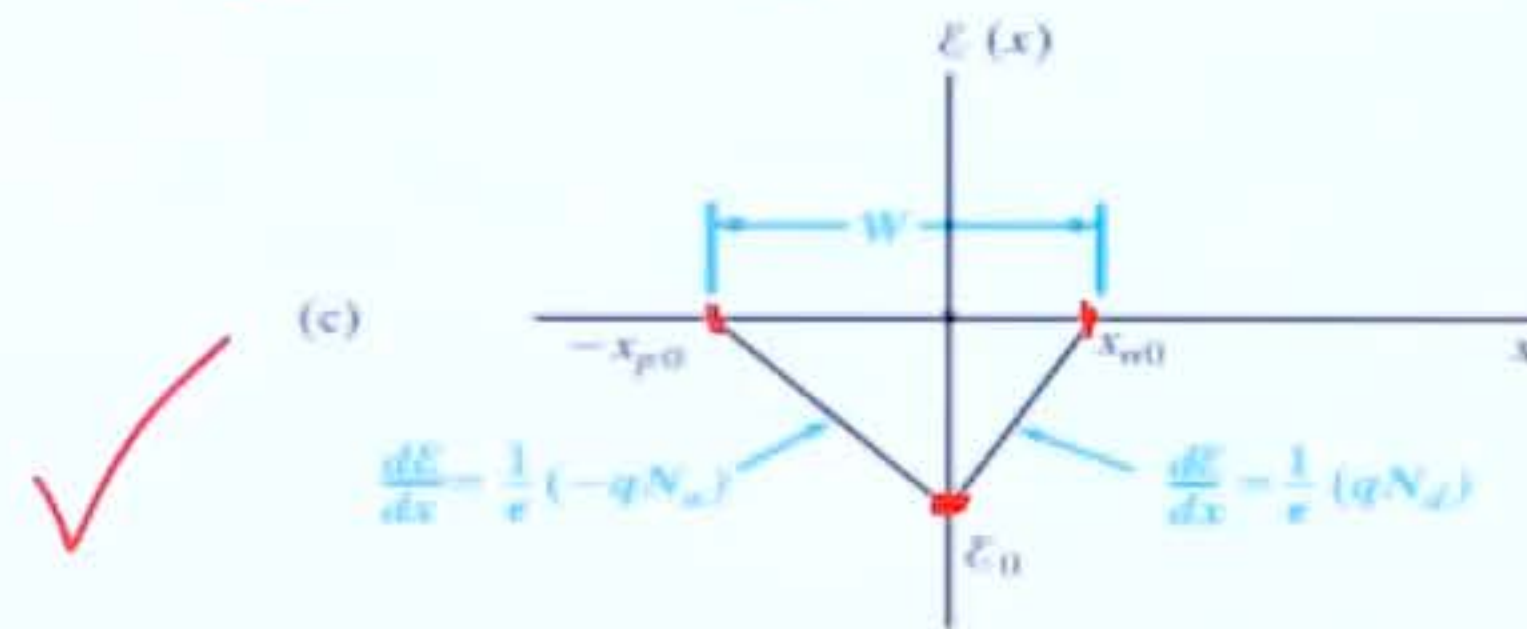
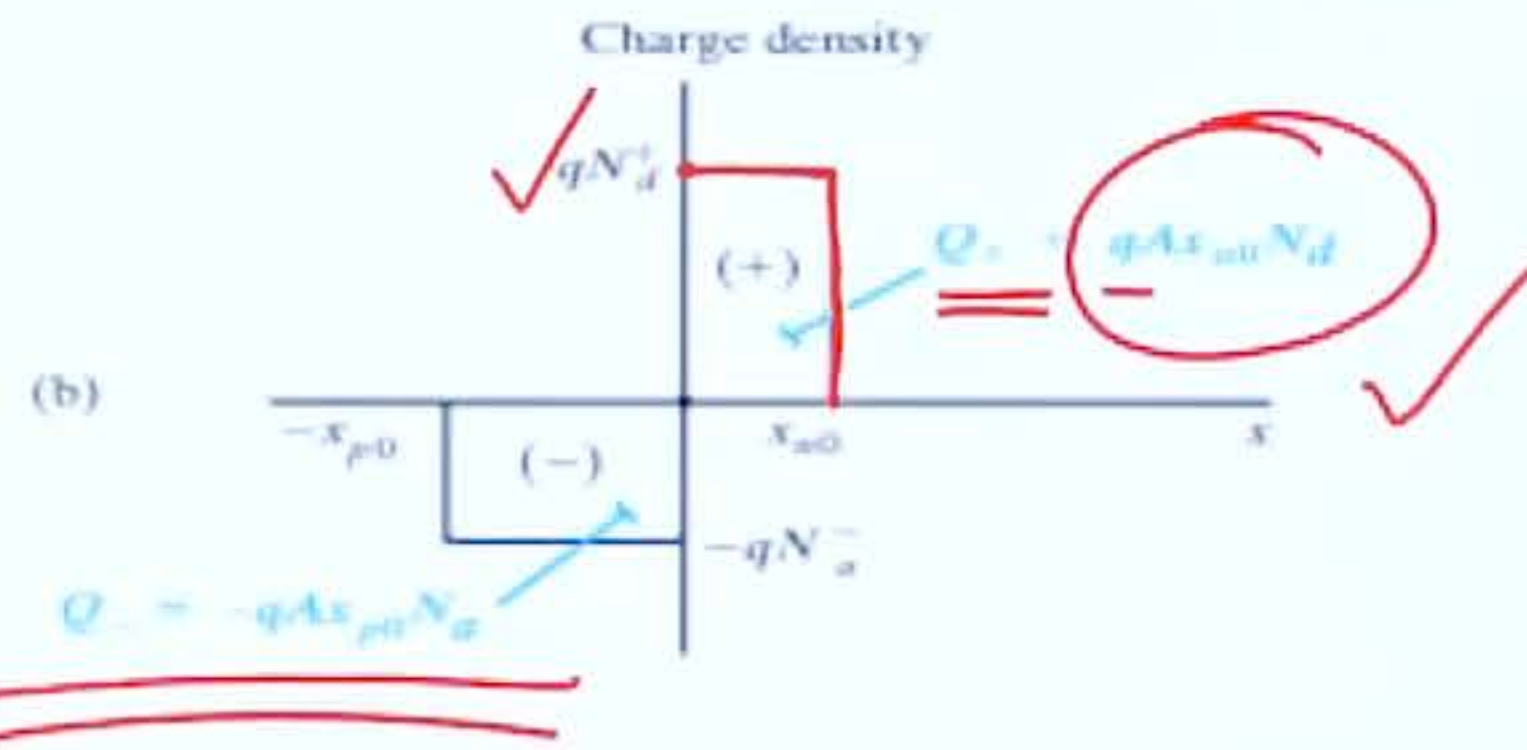
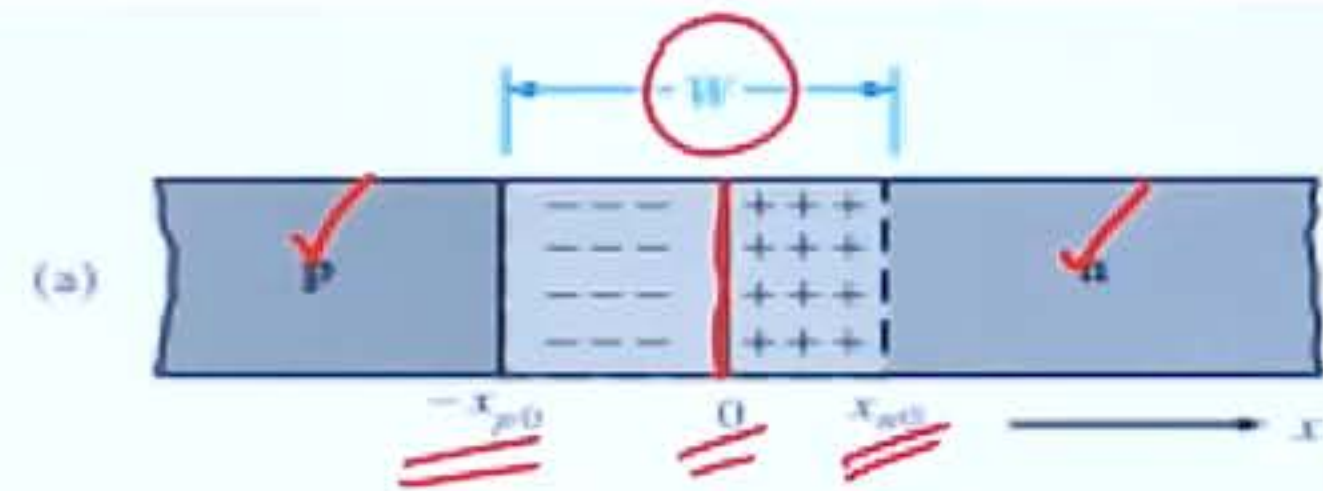
(d) under R.B

(a) Derive the expression for Built in potential of an abrupt PN junction at equilibrium. (7)

(b) An abrupt silicon PN junction has $N_A=10^{17} \text{ cm}^{-3}$ on the p-side and $N_D=10^{15} \text{ cm}^{-3}$ on the n-side. The area of cross section of the diode is 10^{-4} cm^2 . The relative permittivity of Si is 11.8. Calculate the built in voltage (V_0) and depletion layer width (W_0) at 300K. (7)

16 a)

V_0



Total charge on both sides

$$\cancel{qAx_pN_a} = \cancel{qAx_nN_d}$$

To calculate the electric field distribution within the transition region, we begin with Poisson's equation, which relates the gradient of the electric field to the local space charge at any point x :

$$\checkmark \frac{d\mathcal{E}(x)}{dx} = \frac{q}{\epsilon}(p - n + N_d^+ - N_a^-)$$

$$\frac{d\mathcal{E}}{dx} = \frac{q}{\epsilon} N_d$$

$$0 < x < x_{m0}$$

$$\frac{d\mathcal{E}}{dx} = -\frac{q}{\epsilon} N_a$$

$$-x_{p0} < x < 0$$

$$\int_{\mathcal{E}_a}^0 d\mathcal{E} = \frac{q}{\epsilon} N_d \int_0^{x_{m0}} dx,$$

$$0 < x < x_{m0}$$

$$\int_0^{\mathcal{E}_0} d\mathcal{E} = -\frac{q}{\epsilon} N_a \int_{-x_{p0}}^0 dx,$$

$$-x_{p0} < x < 0$$

Therefore, the maximum value of the electric field is

$$\mathcal{E}_0 = -\frac{q}{\epsilon} N_d x_{n0} = -\frac{q}{\epsilon} N_a x_{p0}$$

It is simple to relate the electric field to the contact potential V_0 , since the $\mathcal{E}$ field at any x is the negative of the potential gradient at that point:

$$\mathcal{E}(x) = -\frac{dV(x)}{dx} \quad \text{or} \quad -V_0 = \int_{-x_{p0}}^{x_{n0}} \mathcal{E}(x) dx$$

Thus the negative of the contact potential is simply the area under the $\mathcal{E}(x)$ vs. x triangle. This relates the contact potential to the width of the depletion region:

$$V_0 = -\frac{1}{2} \mathcal{E}_0 W = \frac{1}{2} \frac{q}{\epsilon} N_d x_{n0} W$$

$$V_0 = \frac{KT}{q} \ln\left(\frac{N_a N_d}{n_i^2}\right)$$

Since the balance of charge requirement is $x_{n0} N_d = x_{p0} N_a$, and W is simply $x_{p0} + x_{n0}$, we can write $x_{n0} = W N_a / (N_a + N_d)$ in Eq. (5-19):

$$V_0 = \frac{1}{2} \frac{q}{\epsilon} \frac{N_a N_d}{N_a + N_d} W^2$$

16 b)

$\checkmark N_A = 10^{17}$

$\checkmark N_D = 10^{15}$

$V_0 = ?$

$W = ?$

$n_i = 1.5 \times 10^{10}$

$\Sigma_v = 11.8$

$A = 10^{-4} \text{ cm}^2$

$$W = \left[\frac{2\epsilon V_0}{q} \left(\frac{N_a + N_d}{N_a N_d} \right) \right]^{1/2} = \left[\frac{2\epsilon V_0}{q} \left(\frac{1}{N_a} + \frac{1}{N_d} \right) \right]^{1/2}$$

W

$V_0 = \frac{kT}{q} \ln \left(\frac{N_A N_D}{n_i^2} \right)$
 $\nearrow 25 \text{ mV}$